

Digital Asset Landscape 2025

TOTAL MARKET CAPITALIZATION



\$4+ Trillion

First time crossing this threshold

ACTIVE MONTHLY USERS



40-70 Million

+10M YoY growth in 2025

TRANSACTION THROUGHPUT



3,400 TPS

100x growth in 5 years

MONTHLY STABLECOIN VOLUME



\$1.25 Trillion

September 2025 (adjusted volume)

Annual Payment Volume Comparison (Adjusted, \$ Trillions)



"Crypto went mainstream in 2024... more people are using crypto services than ever before."

— a16z State of Crypto 2025

Why Digital Assets Matter to Traditional Finance

Major Financial Institutions Now Onboard
Visa, BlackRock, Fidelity, JPMorgan, and Mastercard all offering crypto products directly to consumers

Real-World Asset (RWA) Tokenization
\$30-34B market in 2025, nearly 4x growth in two years
BlackRock BUIDL fund: \$2.9B in tokenized U.S. Treasuries

Regulatory Clarity Emerging
GENIUS Act enacted for stablecoins; CLARITY Act advancing through Congress
SEC approval process streamlined (270 → 75 days)

Business Value Proposition
24/7 programmable settlement, new yield sources, global distribution, reduced counterparty risk

"DeFi and TradFi must work together to realize the benefits of both" — IMF, 2022

Institutional Adoption Drivers



What's Driving Institutional Adoption

Regulatory Clarity

GENIUS Act (stablecoins) enacted; CLARITY Act advanced
SEC generic listing standards streamline approvals (270→75 days)

Proven Custody Infrastructure

Institutional-grade insurance coverage and SLAs
SOC2/ISO compliance, segregated accounts, audit trails

Yield Opportunities

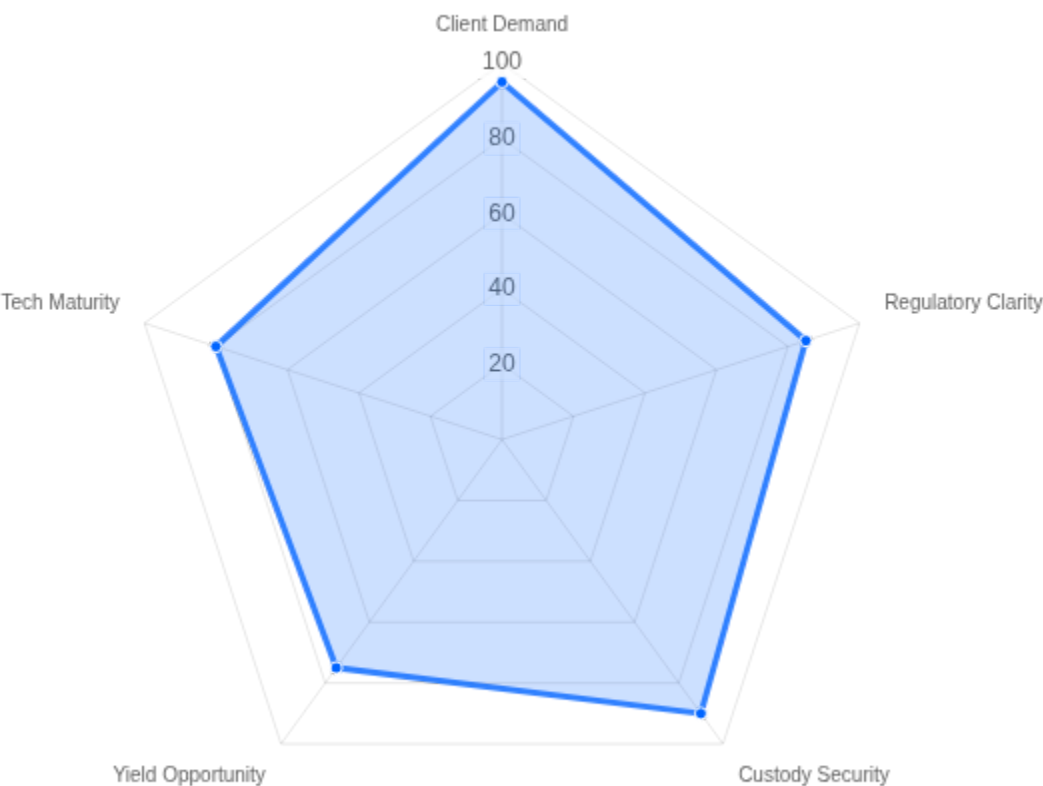
Ethereum staking yields (3-5%); tokenized T-bills
BlackRock BUIDL fund ~\$2.5B AUM, tokenized treasury products

Client Demand

ETPs exceed \$175B in assets (169% YoY growth)
Competitive pressure to offer digital asset exposure

"Stablecoins and blockchain are becoming the backbone of finance" — Forbes, November 2025

Adoption Driver Influence



BlackRock

JPMorgan

Fidelity

Citi

Blockchain Basics

📖 What is Blockchain?

A distributed digital ledger technology that records transactions across many computers to ensure data cannot be altered retroactively

⚙️ How It Works

Transactions are grouped into blocks → Each block contains a cryptographic hash of the previous block → Blocks form an immutable chain

Consensus mechanisms (PoW, PoS) secure the network by validating transactions

🔑 Key Characteristics

Decentralization: No single point of control or failure

Immutability: Once recorded, data cannot be altered

Transparency: All network participants can verify transactions

Security: Cryptographic principles protect data integrity

🏛️ Architectural Benefits for TradFi

Reduces settlement times from T+2 to near-instant

Eliminates reconciliation processes and reduces errors

Creates programmable financial assets and automated compliance

Blockchain Structure Visualization

Blockchain: Chained Blocks with Hash Pointers

Each new block references the previous block's hash, creating an immutable chain



Key Blockchain Properties:

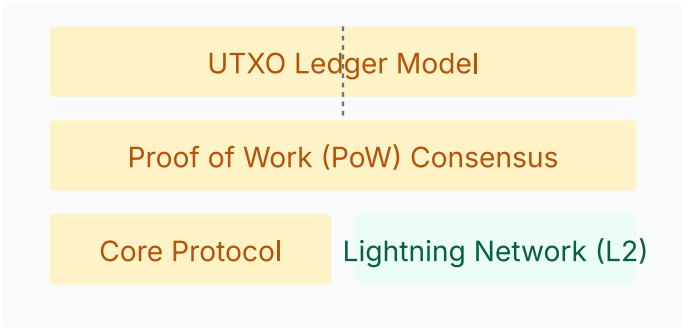
- Immutability: Once a block is added, it cannot be altered without invalidating all subsequent blocks
- Consensus: Network participants agree on the validity of blocks before adding them to the chain

● Each block contains: Timestamp, Transactions, Previous Block Hash, Current Block Hash

● Hash pointers create the chain and ensure immutability

Blockchain Architecture Fundamentals

Bitcoin (BTC)



Model

UTXO (Unspent Transaction Output) model
Simple script language with limited expressivity

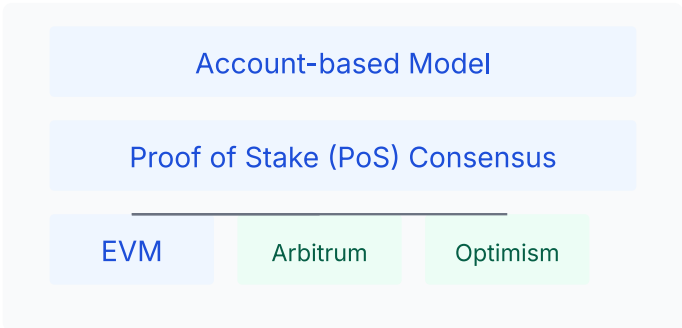
Strengths

High security assurance
Predictable monetary policy
Optimized for store of value (SoV)

Scaling Solution

Layer 2 Lightning Network for payments
Payment channels with near-instant finality

Ethereum (ETH)



Model

Account-based ledger
EVM smart contracts for programmability

Strengths

Programmable state
DeFi and tokenization infrastructure
Developer ecosystem

Scaling Solutions

Rollups: Arbitrum, Base, Optimism
Sub-cent transaction fees on L2s

Speaker notes: UTXO vs account models have direct implications for custody, policy controls, and transaction construction. Ethereum's rollup-centric roadmap makes blockspace elastic for institutional apps.

What is Bitcoin (BTC): Architecture & Why It Matters

UTXO Ledger Model

Unspent Transaction Outputs (UTXOs) model where transactions consume and create outputs; no mutable account balances

Proof-of-Work Consensus

SHA-256 mining secures network; difficulty adjusts every 2016 blocks (~2 weeks); ~10 min block time

Fixed Supply Cap

Hard limit of 21 million BTC; predictable issuance schedule with halvings approximately every 4 years

Layer 2 Scaling

Lightning Network for instant, low-fee payments; state channels with multi-sig contracts

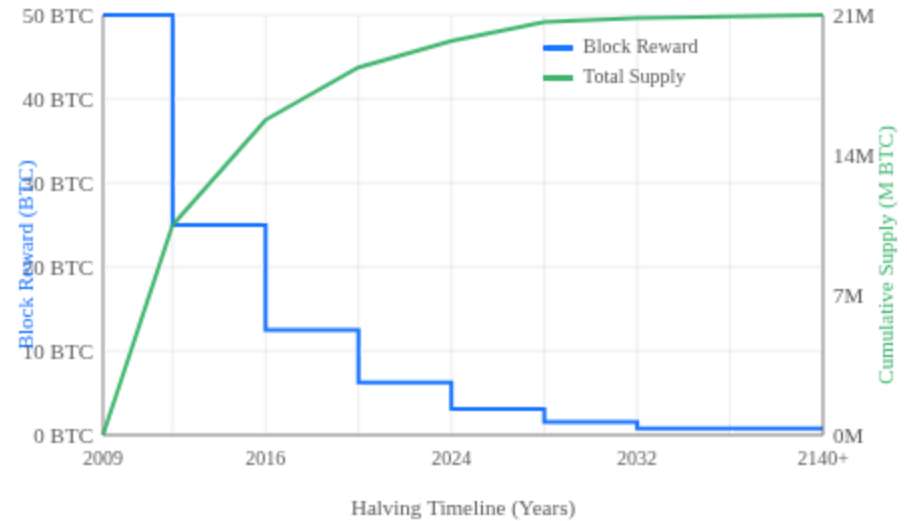
Security Model

Economic cost via hashpower; broad node verification; simple scripting limits attack surface

Institutional Perspective

Digital gold/store of value; high-assurance monetary policy; ETP access; regulatory clarity growing

Bitcoin Supply Issuance & Halving Timeline



Key Halving Events

2009: Genesis (50 BTC)
2012: First halving (25 BTC)
2016: Second halving (12.5 BTC)
2020: Third halving (6.25 BTC)
2024: Fourth halving (3.125 BTC)

Supply Milestones

2022: 19M BTC (90% of total)
2025: 19.5M BTC (93% of total)
2032: 20M BTC (95% of total)
2140: 21M BTC (100% - final)

What is Ethereum (ETH): Programmable Settlement Layer

🔧 General-Purpose Computation Platform

"World computer" enabling programmable smart contracts for DeFi, tokenization, and decentralized applications

📦 Account-Based Model

EOA (user accounts) and Contract Accounts maintain state; fundamentally different from Bitcoin's UTXO model

🔗 Execution Environment (EVM)

Ethereum Virtual Machine runs bytecode compiled from languages like Solidity/Vyper; gas metering for execution

📦 Proof-of-Stake Consensus

Post-Merge (2022): Validators stake 32 ETH, propose/attest blocks, slashing for violations; reduced energy use by ~99.95%

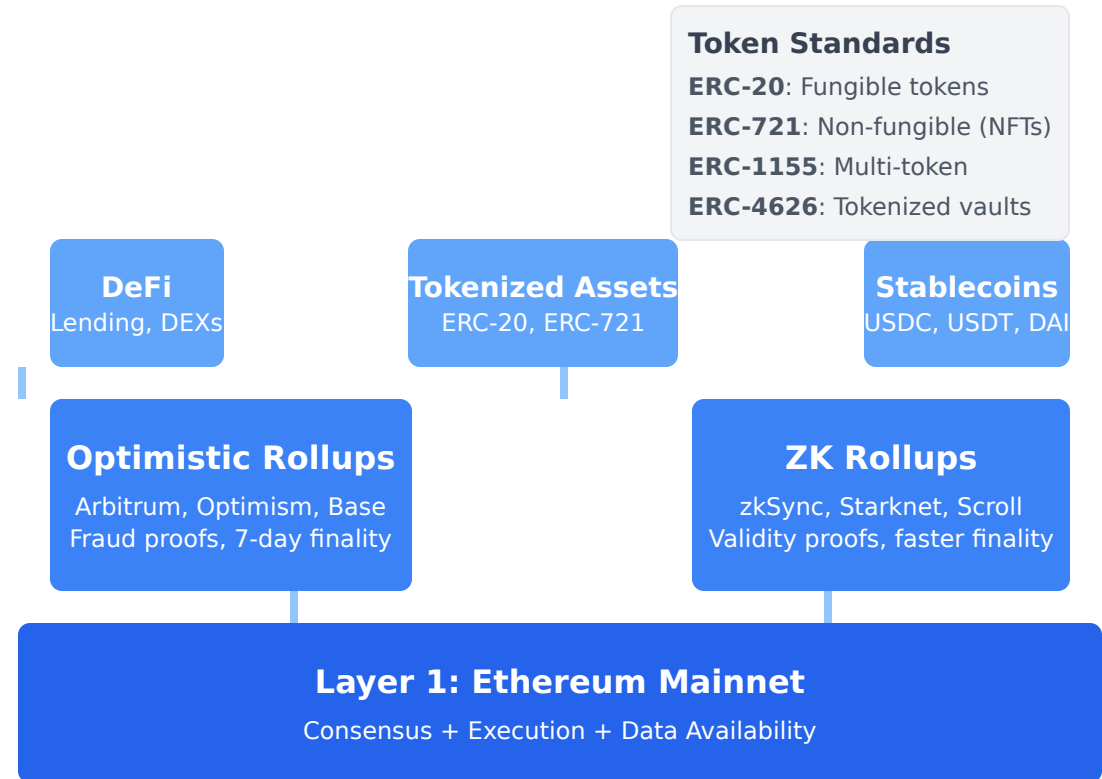
📦 Scaling via Layer 2 Solutions

Optimistic Rollups: Arbitrum, Optimism, Base; ZK Rollups: zkSync, Starknet, Scroll; fees < \$0.01 vs L1 (\$2-20)

🔌 Institutional Applications

Tokenized assets, stablecoin settlement, permissioned DeFi, Ethereum staking (3-5% yield), compliance layers

Ethereum L1/L2 Architecture Stack



Institutional considerations: Smart contract risk, MEV (Maximum Extractable Value), L2 bridge security, regulatory framework

UTXO vs Account Model

Architectural Models for Transaction & State Management

UTXO Model

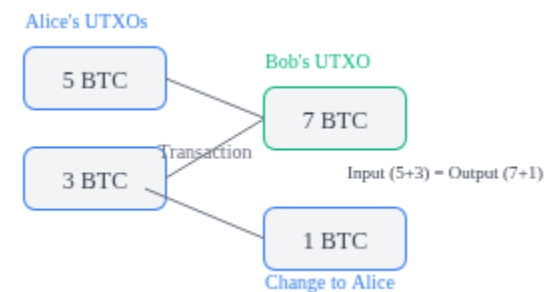


Unspent Transaction Output - Used by Bitcoin

What It Is

- Tracks tokens as discrete units (like cash/coins)
- No accounts - only unspent outputs from previous transactions

Transaction Example



Alice spends 2 UTXOs (5 + 3 BTC) to send Bob 7 BTC, receiving 1 BTC change

Privacy

- Better pseudonymity through new addresses for each transaction

Scalability

- Allows parallel transaction processing (higher throughput)

Security

- Built-in double-spend protection - UTXOs can only be spent once

Complexity

- More complex for wallet management and smart contracts

Real-world analogy: Like physical cash where you must use whole bills and receive change

Account Model

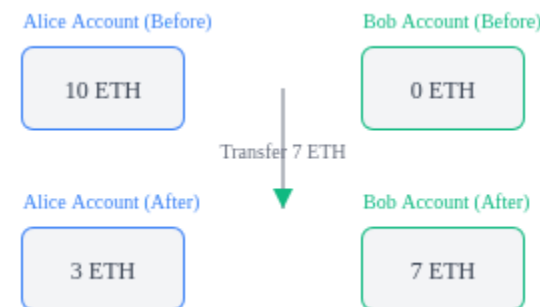


Balance-Based Ledger - Used by Ethereum

What It Is

- Maintains a single balance per account
- Direct debit/credit system (like a bank account)

Transaction Example



Alice's account balance of 10 ETH is reduced by 7 ETH, Bob's increases by 7 ETH

Privacy

- Transactions linked to persistent account identities (less private)

Scalability

- Sequential processing required (potential throughput bottleneck)

Security

- Uses account nonces and state verification to prevent double-spends

Complexity

- Simpler for developers and users, better for smart contracts

Real-world analogy: Like a bank account with balance that increases and decreases

Architectural Impact

The choice between UTXO and Account models affects custody implementation, smart contract design, privacy controls, and transaction construction patterns for institutional applications.

Token Taxonomy & Lifecycle

Definition
Onchain representation of value/rights/utility governed by smart contracts

- Token Types**
- Fungible (ERC-20):** Identical units for payments, governance, RWAs
 - Non-fungible (ERC-721):** Unique items (collectibles, credentials)
 - Semi-fungible (ERC-1155):** Batched IDs for gaming/commerce

- Functional Categories**
- Utility:** Access rights, platform fees, services
 - Governance:** Voting power, proposal submission
 - Financial:** Yield-bearing assets, RWAs, dividends
 - Compliance:** Permissioned transfers with KYC

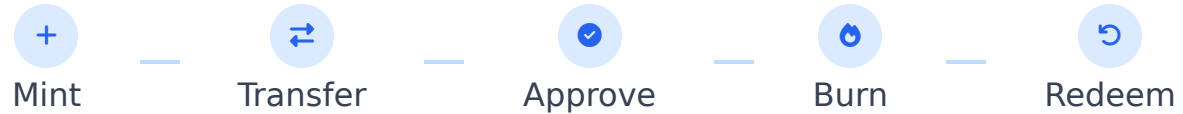
- Key Standards**
- ERC-20: Fungible tokens (transfer, balanceOf)
 - ERC-721: NFTs (ownerOf, tokenURI)
 - ERC-4626: Tokenized vaults (deposit, withdraw)
 - ERC-3643/T-REX: Compliance frameworks

Token Taxonomy

Blockchain Tokens

Fungible ERC-20	Non-Fungible ERC-721	Semi-Fungible ERC-1155
Payment Tokens USDC, ETH, BTC	Digital Art Unique ownership	Gaming Assets Items, skins
Utility Tokens Platform access, fees	Identity Verifiable credentials	Event Tickets Batched, time-bound
Governance Voting power, proposals	Real Estate Property titles, fractions	Hybrid Items Mixed fungible/unique

Token Lifecycle



Architectural Impact

Composability enables tokens to be seamlessly integrated across DeFi protocols, treasuries, and payment systems, with policy enforcement through programmable code

What is an RWA (Real-World Asset) Tokenization

Definition

On-chain tokens representing claims on off-chain financial or physical assets

Asset Classes

- Treasuries & Money Market Funds (\$8B+)
- Bonds & Fixed Income (\$12B+)
- Real Estate (\$6B+)
- Private Credit & Invoices (\$4B+)
- Commodities (\$2B+)

Issuance Stack

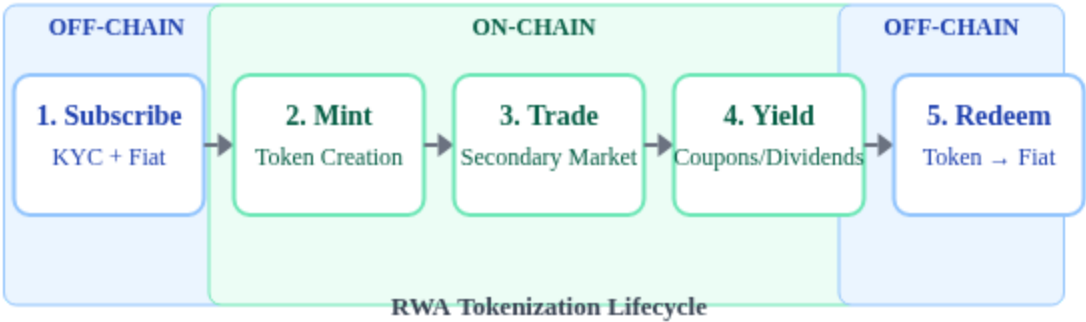
- Sponsor/Issuer + SPV structure
- Transfer agent/registrar with KYC/IDP
- Custodian of underlying assets
- Smart contract (permissioned token)
- Market venues & redemption modules

Benefits & Constraints

- ✓ 24/7 settlement, fractionalization
- ✓ Programmable compliance
- ✓ Global distribution, composability
- ✗ Securities laws, transfer restrictions
- ✗ Oracle/valuation risk

Example: BlackRock BUIDL fund (\$2.5B tokenized U.S. Treasuries)

RWA Lifecycle



Market Size
\$33-36B (2025)
4x growth in 2 years

Industry Adoption
BlackRock, JP Morgan, Ondo
Traditional finance leaders

Real-World Asset (RWA) Tokenization

Market Size & Growth

\$30-34B in 2025, nearly 400% growth in two years
Projected to reach \$16T by 2030 (BCG/Ripple report)

BlackRock BUIDL Fund

\$2.9B in tokenized U.S. Treasuries (largest fund)
Now accepted as collateral across major trading venues

RWA Use Cases

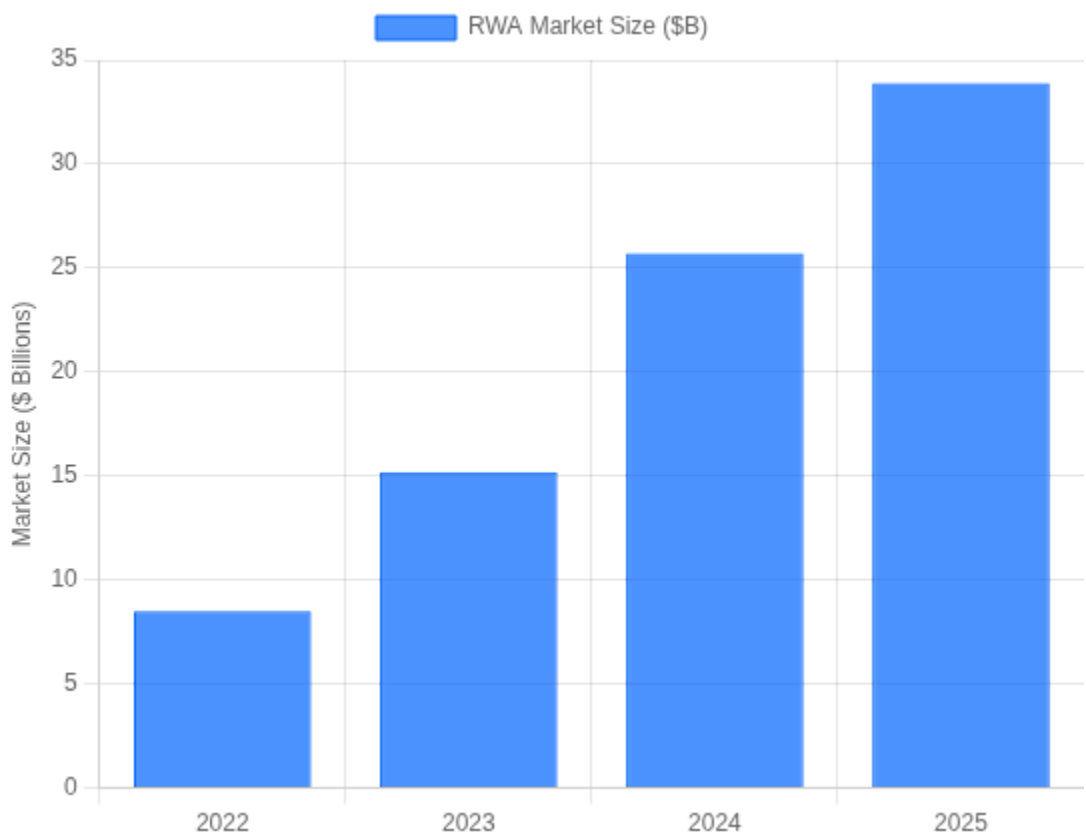
Treasuries & Money Market Funds: 55% of market
Private Credit: 22% | Real Estate: 15% | Commodities: 8%

Key Benefits

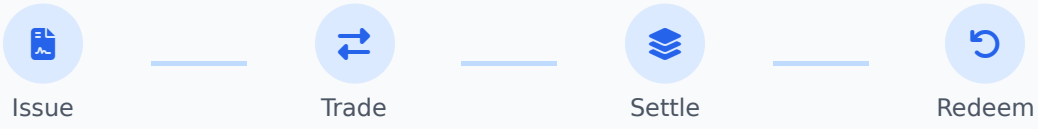
24/7 settlement, fractional ownership, programmable compliance, enhanced liquidity for traditionally illiquid assets

"Tokenization is the future of financial markets" — Larry Fink, BlackRock CEO

RWA Market Growth 2022-2025



RWA Tokenization Lifecycle



What is a Stablecoin: Design, Types, and Institutional Uses



Definition

Cryptoasset designed to maintain a stable value (typically USD-pegged) for payments, settlement, and treasury operations



Types & Mechanics

Fiat-backed: Fully reserved cash/T-bills in banks/Trusts (USDC, USDT); 1:1 mint/redeem via KYC'd issuers

Crypto-collateralized: Over-collateralized with on-chain assets (DAI) + liquidation engines

Algorithmic: Peg via algorithmic supply incentives (higher depeg risk)



Institutional Use Cases

Cross-border settlement (24/7, low-fee, global reach)

Treasury management & sweeps (yield-generating)

On/off-exchange collateral for trading

Marketplace payouts & programmable disbursements



Regulatory Landscape

2025 GENIUS Act establishes stablecoin framework in US

Requirements: licensing, reserve custody, audits/attestations, redemption guarantees

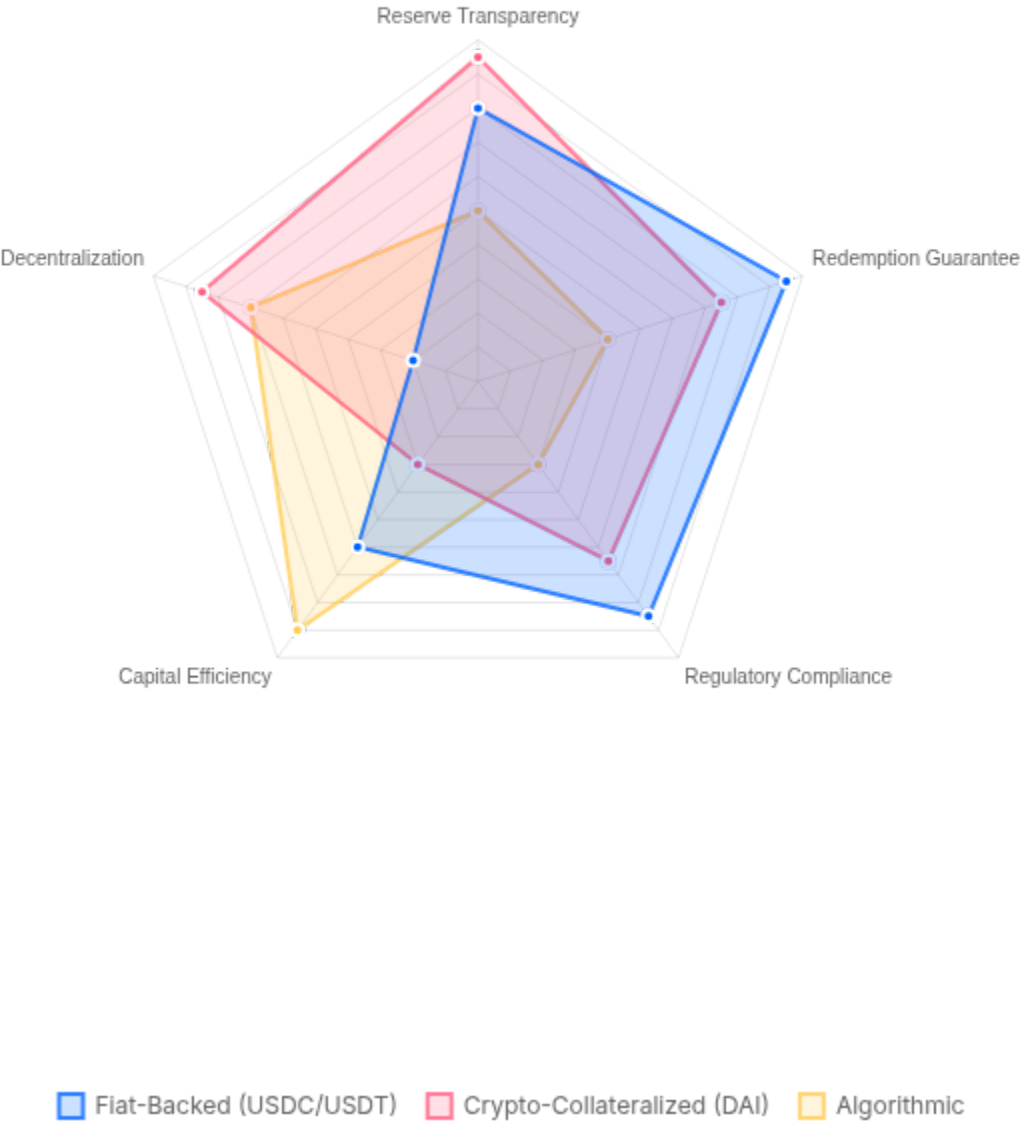


Risk Management

Counterparty/issuer risk, chain/bridge risk, depeg liquidity, sanctions compliance

"Stablecoins processed \$9T in 12 months (87% YoY growth), reaching \$1.25T in September 2025 alone" — a16z State of Crypto 2025

Stablecoin Types: Reserve Backing & Risk Comparison



Mint/Redeem Flow: Fiat-backed Stablecoins



Stablecoins as Financial Infrastructure

ANNUAL ADJUSTED VOLUME



\$9 Trillion

87% YoY growth in 2025

MONTHLY PEAK VOLUME



\$1.25 Trillion

September 2025 (adjusted)

TOTAL SUPPLY



\$300+ Billion

All-time high in 2025

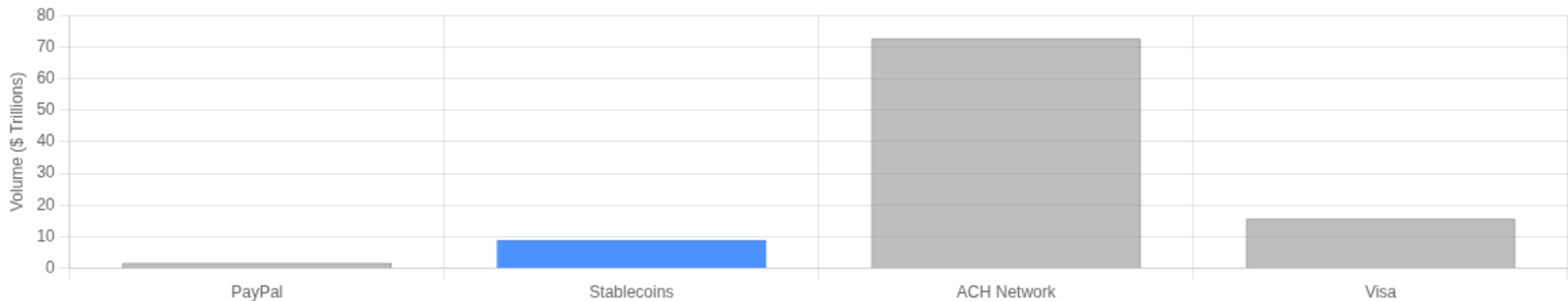
U.S. TREASURY HOLDINGS



#17 Holder

Over \$150B in U.S. Treasuries

Annual Payment Volume Comparison (\$ Trillions)



"Stablecoins and blockchain are becoming the backbone of finance"

— Forbes, November 2025

Cryptocurrency Wallet Types



Software Wallets (Hot/Warm)



Examples

Mobile: Coinbase Wallet, MetaMask, Trust Wallet
Desktop: Electrum, Exodus, Atomic
Web: Web3 wallets, browser extensions

Advantages

High convenience, DeFi protocol integration
Real-time market data and trading

Risks

Online attack surface (phishing, malware)
Requires robust device security

Hardware Wallets (Cold)



Examples

USB devices: Ledger Nano X/S, Trezor Model T
Air-gapped: Coldcard, Keystone, BitBox
HSM: Institutional-grade hardware modules

Advantages

Highest isolation from network threats
Private keys never leave the device

Tradeoffs

Lower operational agility, physical security needs
Multi-step process for transactions

Institutional Implementation Examples

Coinbase

Hot (2%): API-driven trading engine with MPC
Warm (3%): Time-locked with 24-48h delays
Cold (95%): Geographically distributed HSMs
Security: SOC2 Type II, \$320M insurance

Fireblocks

MPC-CMP: Threshold signature scheme (TSS)
Policy Engine: Chain-agnostic approval rules
SGX Enclaves: Hardware-secured environment
Clients: BNY Mellon, Six Digital Exchange

Exchanges (Binance/Kraken)

Binance SAFU: 10% trading fees to secure fund
Hot: ~2% of assets for immediate withdrawals
Kraken: Air-gapped cold storage with global distribution
Security: Quarterly proof-of-reserve audits

Licensing & Compliance

NY BitLicense (Coinbase)
OCC Charter (Anchorage)

Transaction Latency

Hot: Seconds
Warm: Minutes/Hours
Cold: Hours/Days

Security Controls

Geo-distribution
Time locks
Quorum approvals

Insurance Coverage

Coinbase: \$320M
Fireblocks: \$550M
BitGo: \$700M

Speaker notes: Institutional wallets follow defense-in-depth approach: Coinbase keeps 98% in cold storage after 2019 security audit. Fireblocks leverages MPC-CMP to eliminate single points of compromise. Exchange wallets focus on segregated customer assets and insurance. Licensing affects custody requirements.

Custody Models Overview

Self-custody



⚡ Latency

Immediate transaction execution
No approval delays (unless multi-sig)

🛡️ Security Risks

Private key loss or theft (catastrophic)
Social engineering attacks
Inadequate backup procedures

📄 Licensing

No licensing requirements
Non-custodial software providers exempt

☂️ Insurance

Limited or no coverage available
Some hardware wallet theft protection

Third-party custodial



⚡ Latency

Approval delays (minutes to hours)
Tiered approval thresholds based on amount

🛡️ Security Risks

Counterparty/custody risk (insolvency)
Operational risk (internal fraud)
Shared infrastructure risks

📄 Licensing

BitLicense (NYDFS) for NY operations
OCC Charter (Anchorage, Protego)
State Trust Charter (Coinbase, BitGo)

☂️ Insurance

Crime insurance (typically \$100M-\$750M)
SOC1/SOC2 audits and certifications

Speaker notes: Institutions typically choose based on regulatory requirements and operational capabilities. Anchorage has OCC federal charter; Coinbase/BitGo have state trust charters. Self-custody requires robust internal controls while custodial solutions trade convenience for counterparty risk. MPC solutions with Fireblocks/Copper provide middle ground with distributed security.

Multi-Signature (Multi-Sig) Wallets

🔒 M-of-N Signature Architecture

Requires multiple signatures to approve transactions (e.g., 2-of-3, 3-of-5)

Each participant holds a complete private key

🛡️ Benefits

- Eliminates single point of failure
- Transparent on-chain governance
- Battle-tested for DAO treasuries

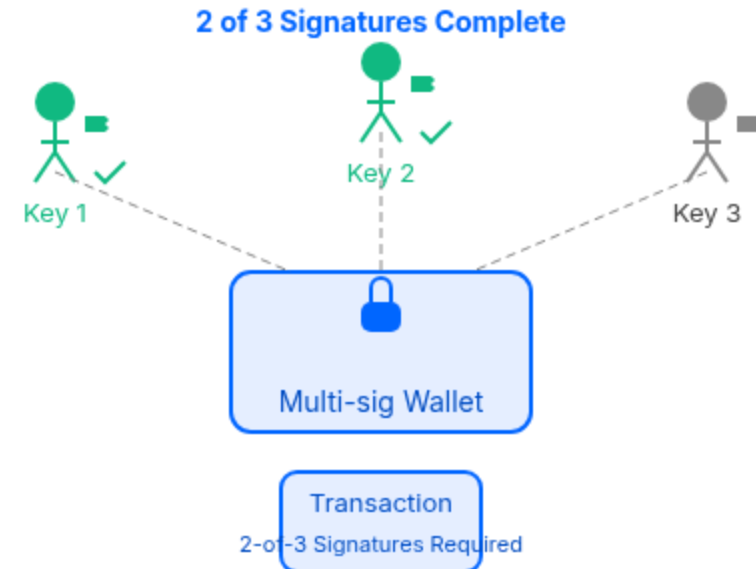
⚠️ Limitations

- Blockchain-specific implementations
- Rigid policies tied to wallet address
- Higher gas fees for multi-sig scripts/contracts

🕒 Historical Vulnerabilities

- Parity Multi-sig Wallet: \$30M stolen due to implementation flaw
- Parity Wallet (Second Incident): \$300M frozen

2-of-3 Multi-Signature Transaction Flow



Reconfiguring multi-sig policy requires creating a new wallet address
Audit tip: Thoroughly verify multi-sig implementation code and establish formal verification procedures

Multi-Party Computation (MPC) Wallets

🔑 Distributed Key Technology

Private key never exists as a whole — split into cryptographic shares using Threshold Signature Scheme (TSS)
Each party holds a key share; key is never reconstructed during wallet creation or signing

🛡️ Key Benefits

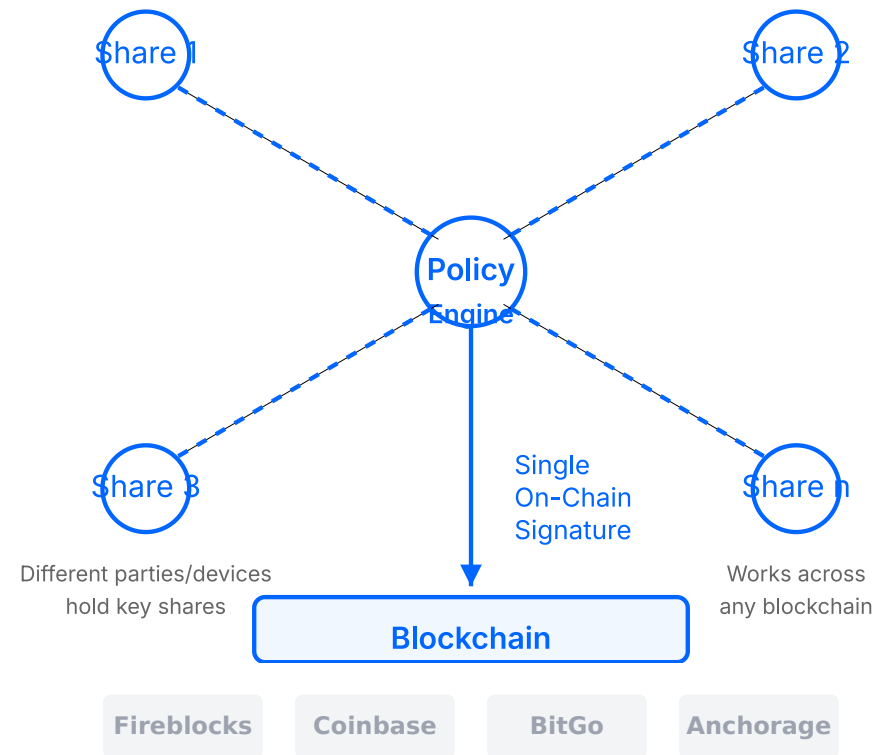
Chain-agnostic (same infrastructure across BTC, ETH, Solana)
Dynamic policies with off-chain signing → lower fees
Enhanced security: no single point of compromise

🏢 Institutional Adoption

Core custody architecture for Fireblocks, Coinbase Prime, BitGo
Powers exchanges and financial institutions managing billions in assets

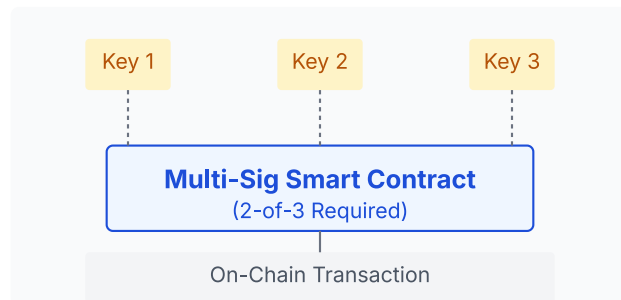
"MPC removes the single point of compromise from both external hackers and insiders as the private key is never concentrated on a single device at any point in time." — Fireblocks

Threshold Signature Scheme (TSS) Architecture



MPC vs Multi-Sig: Architectural Differences

Multi-Sig



Keys & Signatures

- Multiple complete private keys
- Signatures recorded on-chain

Policy & Upgrades

- Policy fixed to wallet address
- Reconfiguration requires migration

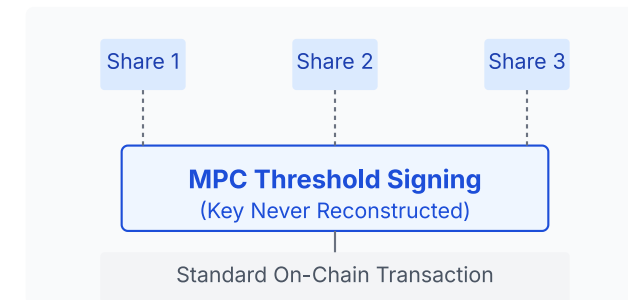
Chain Support & Cost

- Protocol-specific implementations
- Higher gas fees for multi-sig scripts

Governance

- Transparent approvals on-chain
- Excellent for DAO treasuries

MPC



Keys & Signatures

- Single logical key split into shares
- Single on-chain signature artifact

Policy & Upgrades

- Dynamic policies managed off-chain
- No address changes for policy updates

Chain Support & Cost

- Chain-agnostic implementation
- Standard transaction fees (off-chain signing)

Governance

- Off-chain approvals with audit logs
- Ideal for institutional operations

■ Speaker notes: Many institutions blend them: MPC for operational custody; multi-sig for protocol treasuries where on-chain signaling matters.

MPC Vendor Analysis: Institutional Custody Solutions

Criteria	 Fireblocks	 Coinbase Prime	 Anchorage Digital	 BitGo
Security Architecture	MPC-TSS with off-chain signing; hardware isolation; geo-distributed key shares	Hybrid MPC + institutional cold storage; policy controls; segregated accounts	MPC with hardware isolation; role-based governance; audited controls	Battle-tested multi-sig; newer MPC/TSS options; policy engine
Multi-chain Support	Broad coverage: BTC, ETH/L2s, Solana; strong DeFi connectivity; 30+ chains	Wide token support across BTC/EVM/L2s; staking for select assets; 200+ assets	Major L1s/L2s with staking/governance support; institutional-focused	700+ assets/tokens; WBTC admin; broadest coverage overall
Licensing Status	Not a qualified custodian; tech platform for clients' custody operations	NYDFS trust charter; qualified custodian; extensive US compliance stack	Federally chartered OCC bank; qualified custodian with highest regulatory status	SD & NY trust charters; qualified custodian in multiple jurisdictions
Insurance	\$1 billion insurance coverage for digital asset custody; enterprise-grade protection	Commercial crime policy; standard coverage with limits for all clients	Crime/specie policies; details customized by client mandate	\$250 million insurance coverage; qualified custody protection
Transaction Latency	Sub-second to minutes with policy-driven approvals; high operational agility	Warm/cold tiers; batched risk-checked withdrawals; SLA-based timing	API-driven with strong governance approvals; institutional workflows	Warm/hot with policy approvals; cold storage for bulk holdings
Key Differentiators	Chain-agnostic MPC, policy engine, DeFi access at scale; settlement network	Deep liquidity/prime services; ETP/asset manager integrations; exchange lineage	US federal bank charter; strongest regulatory positioning; governance focus	Multisig heritage + MPC flexibility; broadest asset coverage; longest track record
Notable Clients	BNY Mellon, Revolut, BlockFi, Galaxy Digital, Three Arrows Capital	BlackRock, Grayscale, ARK Invest, multiple Bitcoin ETFs, Abrdn	Visa, Apollo, a16z crypto, Wellington, PayPal	Goldman Sachs, Bitstamp, Pantera Capital, Swissborg, Genesis

 **Architect guidance:** Choose tech-only MPC (Fireblocks) for in-house ops; Prime/Anchorage/BitGo for qualified custody needs. Validate SLAs, disaster recovery, incident response, and jurisdictional fit for your specific use case.

Security Exploits: Technical Breakdown & Lessons for Architects

⚠ Mt. Gox (2014, ~\$450M)

Vector: Hot wallet private key compromise over extended period; lack of proper segregation and reconciliation

Root causes: Single-key control; no multi-party approvals; weak accounting controls; inadequate monitoring

Technical impact: Transaction malleability exploited; private keys extracted from unencrypted wallet.dat files

⚠ Binance (2019, ~\$40M)

Vector: Sophisticated phishing/malware campaign that captured API keys and 2FA tokens; coordinated withdrawal from hot wallet

Root causes: Compromised user/API credentials; insufficient withdrawal friction for anomalous patterns

Technical impact: 7,000 BTC exfiltrated in a single transaction bypassing security checks

⚠ Ronin Bridge (2022, ~\$624M)

Vector: Validator key compromise (5-of-9 threshold) including social engineering; attribution to North Korea's Lazarus Group

Root causes: Insufficient decentralization of validator set; weak key custody practices; limited on-chain rate limiting

Technical impact: 173,600 ETH and 25.5M USDC stolen via malicious withdrawals from bridge contract

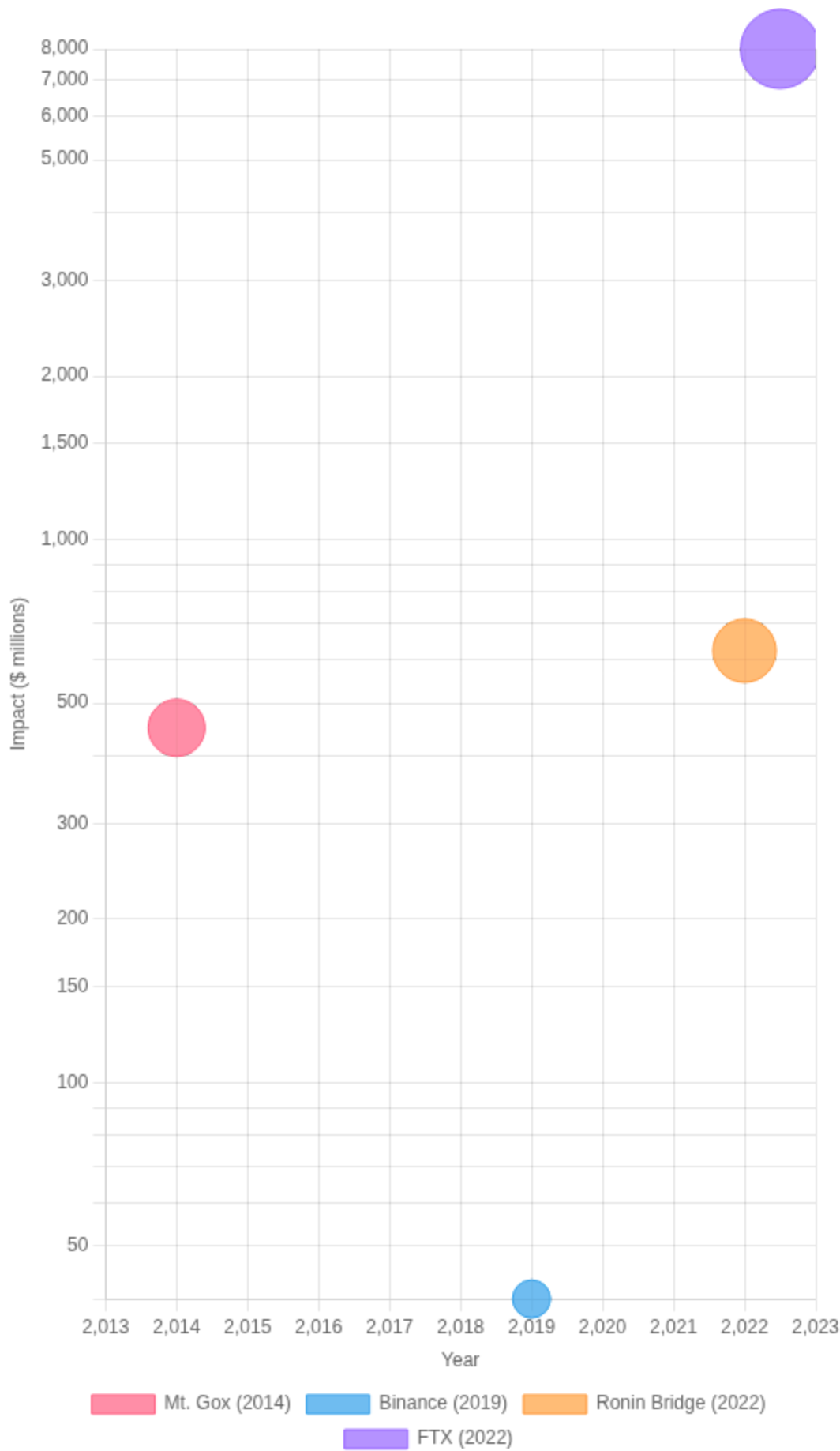
⚠ FTX (2022, ~\$8B governance failure)

Vector: Commingled customer/company funds; inadequate key governance; post-collapse unauthorized transfers

Root causes: Lack of segregation of duties, independent custody, board oversight, and internal controls

Technical impact: Alameda Research accessed FTX customer deposits via shared Fireblocks infrastructure

Impact Analysis & Architectural Lessons



Architect Security Checklist:

Multi-Party Controls

MPC/multisig everywhere with hardware roots of trust (HSM/TEE)

Tiered Storage

Strict velocity limits; just-in-time credentials; device attestation

Anomaly Detection

Automated circuit breakers; rehearsed incident response runbooks

Bridging Security

Decentralized validator sets; threshold upgrades; formal verification

Security Architecture Best Practices

Tiered Storage Architecture

80-90% cold storage (offline, air-gapped)
Warm storage with time-delayed withdrawals
Minimal hot wallets with velocity limits

Multi-Party Controls

MPC or multi-sig for all significant transfers
Dual control for policy and parameter changes

Hardware Security Integration

Hardware Security Modules (HSMs) for key operations
Secure enclaves for cryptographic operations
Device attestation and just-in-time credentials

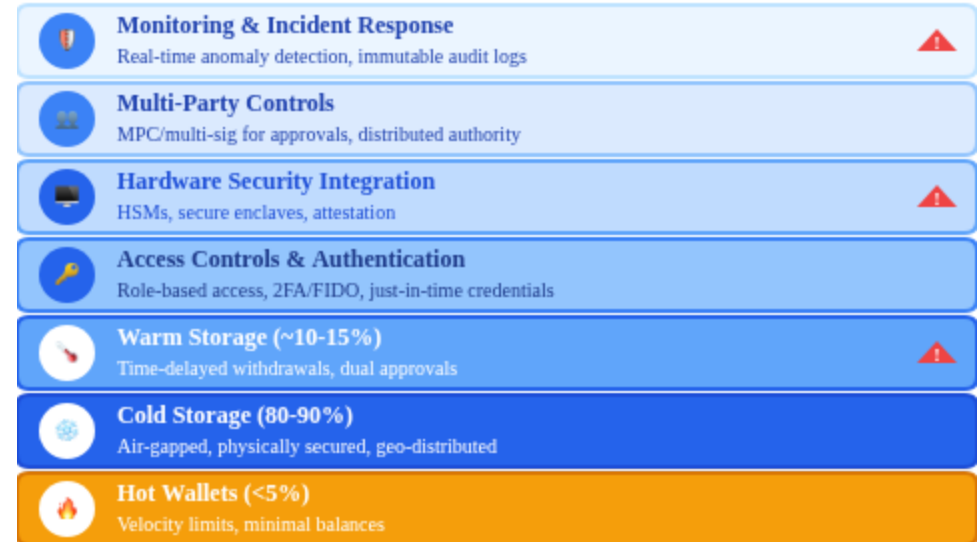
Monitoring & Incident Response

Real-time anomaly detection on transactions
Immutable audit trails and rehearsed response playbooks

Future-Proofing

Post-quantum cryptography roadmap (\$750B BTC vulnerable)
Regular pen-testing and red-team exercises

Defense-in-Depth Security Model



Enterprise-grade security requires multiple protective layers to defend against both external and internal threats